

Wk2/S3 Lecture #6 : DS0OPS

1. Topics Covered

- Understand **binary representation of numbers**
 - Use **bitwise operators**: AND, OR, XOR, NOT
 - Apply **bit masking techniques** to solve problems
 - Identify **unique number problems** using XOR
 - Generate **Power Set** using bit masking
 - Solve **MCQ-based conceptual questions**
 - Write **coding solutions** using bitwise logic
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2. Why Bit Masking?

Bit Masking allows us to:

- Store multiple boolean states inside **one integer**
- Perform **fast operations** ($O(1)$)
- Optimize space and time (used in compilers, OS, networking, cryptography)

Real-Life Analogy

Think of a **switch board**:

- ON = 1
- OFF = 0

Each switch is one **bit**.

3. Binary Representation

Decimal	Binary
---------	--------

1	0001
---	------

2	0010
---	------

3	0011
---	------

4	0100
---	------

5	0101
---	------

Bit positions are counted from **right to left**, starting at **0**.

4. Bitwise Operators

4.1 Bitwise AND (&)

Rule:

`1 & 1 = 1`

`Else = 0`

Use Case:

- Check **odd / even**
- Check if a bit is **set**

```
if ((n & 1) == 1) {  
    // odd  
}
```

4.2 Bitwise OR (|)

Rule:

$0 | 0 = 0$

Else = 1

Use Case:

- Set a bit

```
int mask = 1 << i;  
n = n | mask;
```

4.3 Bitwise XOR (^)

Rule:

$0 \wedge 0 = 0$

$1 \wedge 1 = 0$

$0 \wedge 1 = 1$

$1 \wedge 0 = 1$

Properties:

- $x \wedge x = 0$
- $x \wedge 0 = x$
- XOR is commutative & associative

4.4 Bitwise NOT (~)

Rule:

- Flips all bits

$\sim 5 \rightarrow -(5 + 1) = -6$

5. Left Shift (<<) and Right Shift (>>)

Left Shift

$n \ll k = n \times (2^k)$

```
int x = 5;    // 0101
System.out.println(x << 1); // 1010 = 10
```

Right Shift

$n \gg k = n / (2^k)$

```
int x = 8;    // 1000
System.out.println(x >> 1); // 0100 = 4
```

6. Unique Number Problem

Problem

Every number appears **twice except one**. Find the unique number.

Logic

- Same numbers cancel out ($x \oplus x = 0$)
- Remaining number is the answer

Code

```
int ans = 0;
for (int x : arr) {
    ans ^= x;
}
System.out.println(ans);
```

Dry Run Example

Array: {2, 3, 5, 3, 2}

Step	ans	XOR with	Result
Start	0	–	0
1	0	2	2
2	2	3	1
3	1	5	4
4	4	3	7
5	7	2	5

Output = **5**

7. Power Set Using Bit Masking

For a set of size n

Total subsets = 2^n

Example

Set = {a, b, c}

Binary from 000 → 111

```
for (int i = 0; i < (1 << n); i++) {  
    for (int j = 0; j < n; j++) {  
        if ((i & (1 << j)) != 0) {  
            System.out.print(arr[j]);  
        }  
    }  
    System.out.println();  
}
```

8. Common Bit Tricks

Operation	Formula
Check odd	$n \& 1$
Clear ith bit	$n \& \sim(1 \ll i)$
Toggle ith bit	$n \wedge (1 \ll i)$
Check ith bit	$(n \& (1 \ll i)) \neq 0$

10. MCQs

1. What is the output of $5 \wedge 5$?

0

2. $n \& (n - 1)$ is used to:

- a) Multiply
- b) Remove rightmost set bit
- c) Divide

- d) Toggle bit

b)

3. Total subsets of a set of size 4?

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11. Coding Questions

1. Find unique number when all others appear twice
2. Check if a number is power of 2
3. Generate all subsets using bit masking
4. Toggle the kth bit of a number
5. Count set bits in an integer